

## Module 5 — Synthesis Quiz

[Question bank](#) · [After exercise lab](#) · [open notes](#) · [two attempts](#)

# Module 5, Synthesis quiz

**Readings:** [Read 0](#), [Read 1](#), [Read 2](#)

**Labs:** [Demo](#), [Exercise](#)

**Canvas:** Synthesis quiz (Module 5) · **two attempts**, highest score counts

Full integration for [Project 5](#): read scatterplots, interpret  $r$ , interpret regression output, predict with `predict()`, distinguish CI vs PI, and reflect on the **five-project portfolio**.

---

## Instructions

- Open notes ([methods map](#), [R reference](#)).
  - Use the interpretation templates from Read 1–2.
  - For R items, match the workflow in the [demo lab](#).
- 

## Section A — Reading a scatterplot

### A1. Describe an association

A scatterplot shows fertilizer dose (x) and plant height in cm (y): the points slope upward and cluster tightly near a line. Describe the association using form, direction, and strength.

### A2. Curved vs linear

A scatterplot of age and reading speed shows a clear arc (reading peaks in mid-life and declines at both ends). Is Pearson correlation an appropriate summary? Explain.

## Answer key — Section A — Reading a scatterplot

1. **Form:** roughly linear. **Direction:** positive (more fertilizer  $\rightarrow$  taller plant). **Strength:** strong (points cluster tightly).
  2. **No** — Pearson  $r$  measures linear association only. A curved (non-linear) form makes  $r$  a misleading summary of the relationship.
- 

## Section B — Interpreting correlation and regression output

### B1. Interpret $r$

R gives `cor(exercise_min, resting_hr) = -0.78`. Interpret this value in context.

### B2. Interpret regression output

R output: (Intercept) 79.1, `exercise_min` -0.30. Write the fitted equation and interpret the slope.

### B3. Point prediction

Using the equation above, predict resting HR for someone who exercises 30 min/day.

### B4. Residual computation

A person exercises 30 min/day and has a resting HR of 74 bpm. Using the model above, compute their residual.

## Answer key — Section B — Interpreting correlation and regression output

1. **Strong negative** linear association: as daily exercise minutes increase, resting heart rate tends to decrease. The relationship is strong ( $|r|$  close to 1).
  2.  $\hat{\text{resting HR}} = 79.1 - 0.30(\text{exercise min})$ . **Slope:** each additional minute of daily exercise is associated with a predicted **0.30 bpm decrease** in resting heart rate.
  3.  $\hat{y} = 79.1 - 0.30(30) = 79.1 - 9.0 = 70.1$  bpm.
  4. Residual =  $74 - 70.1 = 3.9$  bpm — they have a resting HR **3.9 bpm higher** than the model predicts.
- 

## Section C — CI for slope

### C1. CI for slope — fertilizer

For the plant model, `confint(fit_plant, 'dose')`  $\rightarrow [1.7, 2.6]$ . Interpret in context.

### C2. Practical significance

The 95% CI for slope is  $[0.001, 0.003]$ , not containing 0. Is this association **statistically significant**? Is it **practically significant**?

### Answer key — Section C — CI for slope

1. We are 95% confident that each additional unit of fertilizer dose changes the mean plant height by between **1.7 and 2.6 cm**.
  2. **Statistically significant** (CI excludes 0). **Practical significance** depends on context — a change of 0.001–0.003 units per 1-unit increase may be too small to matter in practice.
- 

## Section D — Mean-response CI vs prediction interval

### D1. Identify correct interval

A biologist wants to estimate the **average** plant height for all plants given exactly 7 units of fertilizer. Should they use `interval = 'confidence'` or `interval = 'prediction'`?

### D2. Identify correct interval — individual

A researcher wants an interval for **one specific new plant** given 7 units of fertilizer. Which interval?

### D3. Width comparison

At  $x_0 = 7$ : CI for mean =  $[22.8, 25.6]$ ; PI for individual =  $[13.1, 35.3]$ . What accounts for the large difference in width?

## Answer key — Section D — Mean-response CI vs prediction interval

1. `interval = 'confidence'` — this estimates where the **mean response** (average outcome) falls, not an individual plant.
  2. `interval = 'prediction'` — this covers where **one individual** future observation is likely to fall.
  3. The PI adds **individual-to-individual variability** on top of uncertainty in the mean. Even if we knew the mean exactly, individual plants would vary widely around it.
- 

## Section E — Lab situations: Project 5 workflow

### E1. R workflow: fit and predict

Write the R commands (in order) to: (1) fit a regression of `weight_g` on `length_cm` using `stones_dat`, (2) extract the slope CI, (3) predict with a 95% PI at `length_cm = 15`.

### E2. Interpreting `lm()` output

After `summary(fit)`, you see `Estimate` for `length_cm` is 12.4 with `Std. Error` 1.8 and `Pr(>|t|) = 0.0003`. What do these tell you?

### E3. Team data workflow

Your team collected stone weights (g) and widths (cm) in a shared Google Sheet. List the steps to get this data into R and fit a regression.